4. If $f(x)=x^2+3x$ than show that $\Delta f(x)=2x+4$.

5. Evaluate $\Delta\left[\frac{1}{(x+1)(x+2)}\right]$ by taking '1' as the interval of differencing

6. Find the missing entry in the following table

x	0	1	2	3	4
y_x	1	3	9	-	81

7. Following are the population of a district

Year (x)	1881	1891	1901	1911	1921	1931
Population (y) Thousands	363	391	421	-	467	501

Find the population of the year 1911

8. Find the missing entries from the following.

x	0	1	2	3	4	5
$y=f(x)$	0	-	8	15	-	35

5.2 Interpolation

Consider the profit of a manufacturing company in various years as given below:

Year (x)	1986	1987	1988	1990	1991	1992
Profit (Rs. in lakhs)	25	29	24	30	32	31

The profit for the year 1989 is not available .To estimate the profit for 1989 we use the technique called interpolation. Let x denote the year and y denote the profit. The independent variable x is called the argument and the dependent variable y is called the entry. If y is to be estimated for the value of x between two extreme points in a set of values, it is called interpolation.



If y is to be estimated for the values of x which lies outside the given set of the values of it, is called extrapolation.

5.2.1 Methods of interpolation

There are two methods for interpolation. One is Graphical method and the other one is algebraic method.

5.2.2 Graphical method

We are given the 'n' values of x and the corresponding values of y for given function $y=f(x)$. we plot these n observed points $(x_i, y_i), i=1,2,3....$ and draw a free hand curve passing through these plotted points. From the graph so obtained, we can find out the value of y for any intermediate value of x. There is one drawback in the graphic method which states that the value of y obtained is the estimated value of y. The estimated value of y differs from the actual value of y.

Example 5.12

Using graphic method, find the value of y when x = 38 from the following data:

x	10	20	30	40	50	60
y	63	55	44	34	29	22

Solution:

Steps in Graphic method:

Take a suitable scale for the values of x and y, and plot the various points on the graph paper for given values of x and y.

Draw a suitable curve passing through the plotted points.

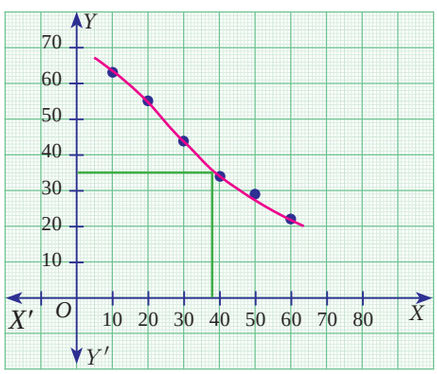


Fig. 5.1

Find the point corresponding to the value x = 38 on the curve and then read the

XII Std - Business Maths & Stat EM Chapter 5.indd 112 2/28/2020 6:09:13 PM